Prince Gnan Praveen Alikana

PhD Researcher | Vision-Language Models | Geospatial AI | Computer Vision and 3D Perception

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Applied machine learning researcher and PhD student at George Mason University building vision-language models for disaster damage assessment across satellite, drone, and ground-level imagery. Pairs research-grade model evaluation with production engineering in LiDAR 3D perception, real-time detection and tracking, and retrieval-augmented LLM systems. Focused on geospatial computer vision and multimodal ML for remote sensing and crisis response.

TECHNICAL SKILLS

Languages	Python, SQL, C++, MATLAB
Computer Vision & 3D Perception	VLM Segmentation (SAM, Mask2Former), YOLO, DeepSORT, LiDAR, SLAM
Geospatial & Earth Observation	Satellite and Aerial Imagery, NASA EO (DSWx, Sentinel-1, IMERG, SMAP)
Machine Learning & Deep Learning	PyTorch, TensorFlow, Hugging Face, Multimodal LLMs, Scikit-learn
Robotics / Systems	ROS2, Linux, Slurm GPU Clusters, Real-Time Inference
LLM / AI Systems	RAG, BERT, FAISS, pgvector, PostgreSQL, Llama 3.3, Whisper
Simulation / Validation	Segmentation Benchmarking (gIoU, cIoU), Simulink, Unreal Engine 5
Cloud & Deployment	Docker, REST APIs, Git/GitHub, Hugging Face Hub, Google Cloud

WORK EXPERIENCE

Graduate Research Assistant – George Mason University, VA. ***July 2026 – Present***

Humanitarian Informatics Lab. NSF-funded research under Dr. Hemant Purohit.

- Benchmarked vision-language segmentation models (LISAT, UniGeoSeg) zero-shot on 10,000+ satellite disaster images on GMU's Hopper GPU cluster, reproducing published results within 0.4 cIoU.
- Uncovered an 8 to 11x accuracy collapse on damage-state queries and 80 to 89% false-positive rates on no-target queries through a split scoring protocol that aggregate metrics miss.
- Extended evaluation from satellite to ground-level imagery across 3 crisis datasets, exposing a 65% vs. 10% accuracy gap between the two architectures.
- **Contributing to EVAC AI, flood evacuation decision support with the City of Virginia Beach, auditing city sensor data and integrating NASA Earth-observation sources.**

Machine Learning Engineer – Community Dreams Foundation. ***April 2026 – July 2026***

- Built a real-time phone voice agent orchestrating Twilio, Whisper STT, Llama 3.3 70B, and Deepgram TTS through LiveKit.
- Designed RAG retrieval on BERT embeddings, FAISS, pgvector, and PostgreSQL, replacing legacy keyword search with semantic retrieval.

Machine Learning Engineer – George Mason University, VA. ***June 2025 – January 2026***

Full-time Machine Learning Engineer delivering externally funded, client-facing AI systems.

- Built LiDAR 3D mapping pipelines (Ouster OS2, ROS2, Open3D, ICP registration, SLAM) from raw sensor capture to aligned 3D maps.
- Developed real-time object detection and multi-object tracking with YOLO, DeepSORT, and ByteTrack over video and ROS2 streams.
- Cut manual validation effort by 60% with MATLAB, Simulink, and Unreal Engine 5 simulation, and sped up GPU training iteration by 45%.

- Containerized inference with Docker and REST APIs for reproducible, client-ready deployment.

Graduate Teaching Assistant – George Mason University, VA. ***August 2023 – May 2025***

- Led graduate labs in SQL optimization, ETL workflows, and scalable data engineering.

Business Data Analyst Intern – Illinois Institute of Technology, IL. ***April 2022 – May 2022***

- Built predictive models in Python, SQL, and Scikit-learn to guide student engagement strategy.

Data Analyst – Vedanta Resources Ltd. ***January 2021 – December 2021***

- Built analytics pipelines on live PLC/SCADA plant data with SQL and Python for engineering teams.

SELECTED PROJECTS & RESEARCH

EVAC AI: Flood Evacuation Decision Support for the City of Virginia Beach

Python | NASA Earth Observation | USGS NWIS | H3 Spatial Indexing

- Evaluated NASA Earth-observation products (DSWx, Sentinel-1, IMERG, SMAP) for coastal flooding, establishing the city gauge network as the primary signal and H3 as the shared spatial index.
- Audited city flood-sensor and infrastructure open data, catching stale sensors and 85 to 96% missing infrastructure fields before they reached the decision pipeline.
- Recommended NOAA tide gauges, the National Water Model, and NWS alerts as real-time sources to cover surge- and tide-driven flooding.

Career Lens: Unlocking Career Potential Through AI Insight

Python | Streamlit | Gemini API | BERT | Random Forest

- Built a resume-analysis platform combining BERT embeddings, Random Forest classification, and Gemini API, reaching 92% skill-gap accuracy and lifting student engagement by 40%.
- Deployed as a Streamlit application with explainable, LLM-generated recommendations for non-technical users.

AI-Driven Predictive Analytics for Early Diabetes Detection

Python | TensorFlow | PyTorch | Big Data

- Modeled early diabetes risk on 100,000+ patient records, improving accuracy by 15% and cutting experimentation time by 40% with GPU-accelerated training.
- Presented results at a university research symposium.

PUBLICATIONS

P. G. P. Alikana, S. P. Deshmukh, and H. Purohit, "What Aggregate Accuracy Hides: A Failure-Mode Analysis of Language-Guided Disaster Segmentation for Mitigation Planning," *AAAI Fall Symposium Series: Planning for a Better Planet*, 2026. (Accepted)

EDUCATION

George Mason University, Fairfax, VA. ***August 2026 – Present***

Ph.D. in Information Technology, College of Engineering and Computing

George Mason University, Fairfax, VA. ***May 2025***

M.S. in Applied Information Technology - Machine Learning Engineering GPA: 3.9/4.0

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar, India. ***2020***

Bachelor of Technology in Instrumentation and Control Engineering